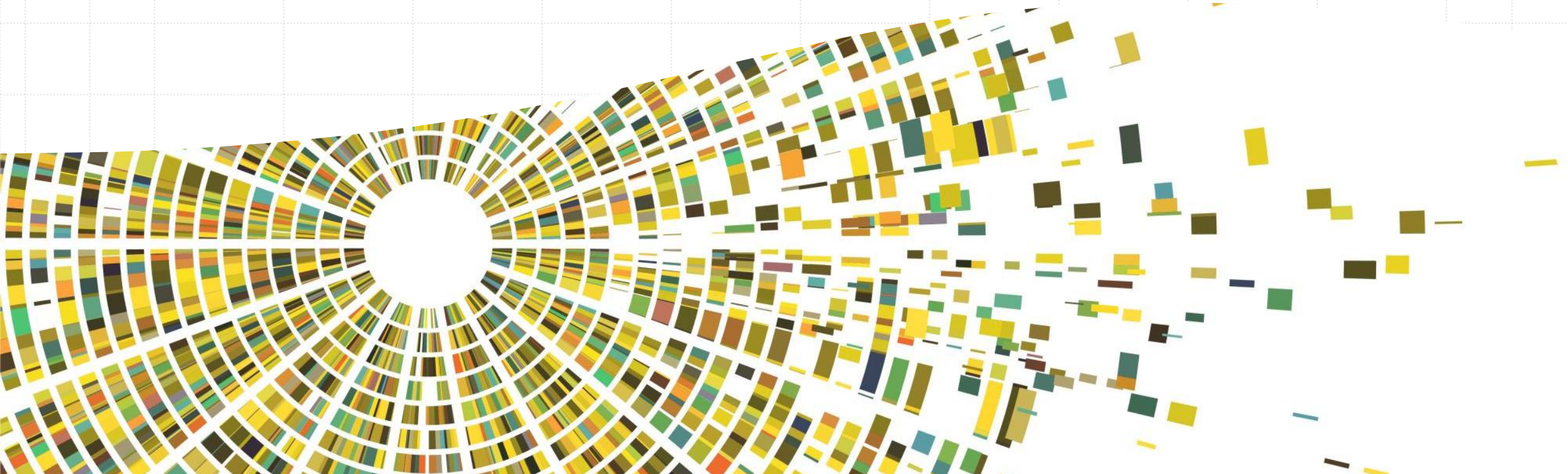


Supercharge Your Agents: Custom Tools, MCP, and Microsoft Foundry

Alvin Ashcraft

Sr. Content Developer – Microsoft



About me



Software industry over 30 years



Morning Dew link blog



Authored three books – Packt Publishing

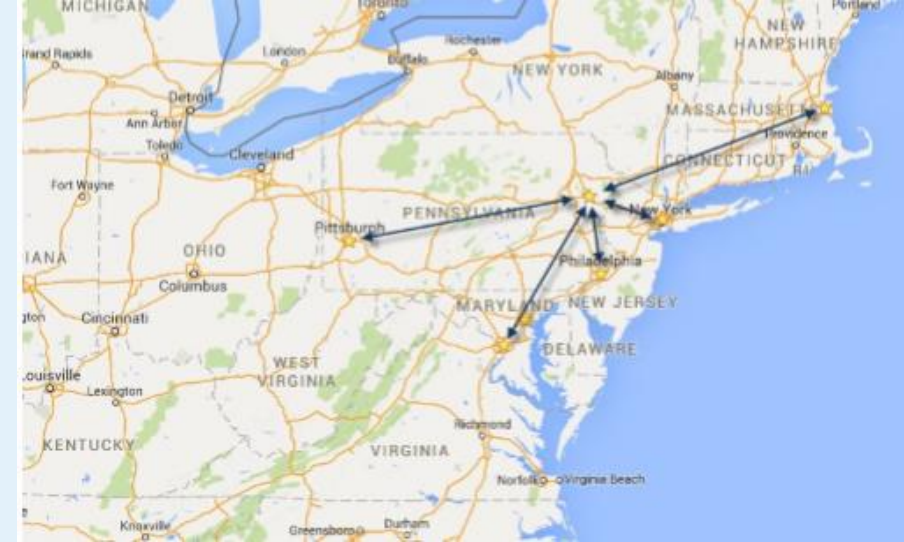


Content developer at Microsoft since 2022



TechBash dev conference organizer since 2015

- Great speakers with top content
- A fraction of the cost of the more crowded conferences
 - 3-day conference plus lodging for ~\$1000
- Full-day deep dive preconference sessions available
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- In addition to the sessions, you get a great hallway Track, amazing food, attendee Welcome Reception, Game Night & more
- Family Day Friday - full day of kids' sessions, free for attendees' families
- Discounted Kalahari Resort rooms with water park access: stay, learn & play all week



<https://techbash.com> or @techbash



Agenda

What is Microsoft Foundry?

Overview of Foundry Agent Services

Hosted Agents in Foundry

Agent Tools

Toolboxes in Foundry

Manage agents at scale

Q&A

What is Microsoft Foundry?



AI services

Agents/models/tools



Enterprise capabilities and
configurations



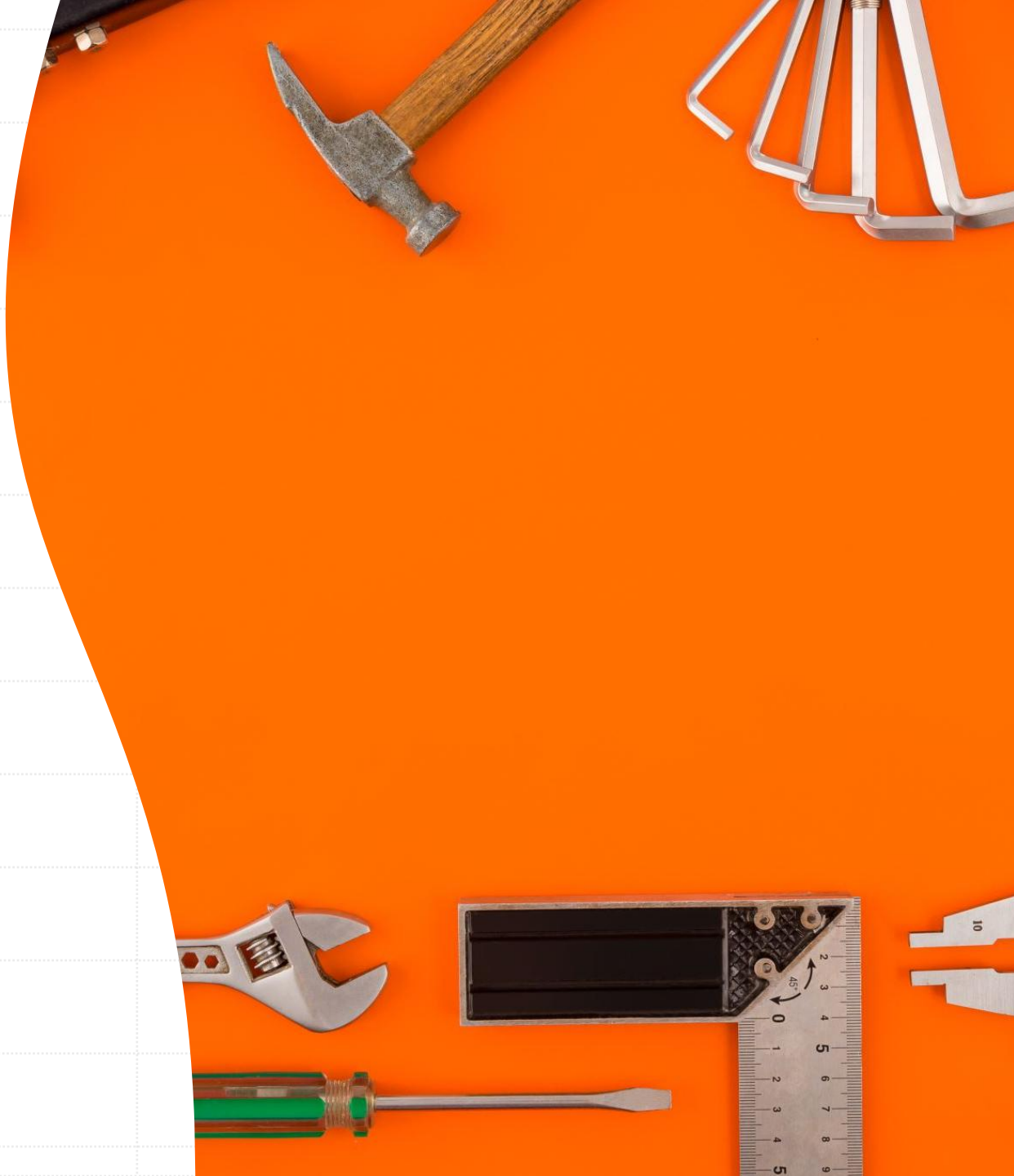
Unified role-based access control
(RBAC) and policies



Networking services

What can you build in Foundry?

- Agents
 - Prompt agents
 - Hosted agents
- Models
 - Nearly 2000 models
 - Microsoft, OpenAI, Anthropic, Meta, Grok, and more
- Tools/knowledge
 - Built-in or custom tools





Agent Services

Agent
Runtime

Responses
API

Publishing

Models

Identity

Security

Observability

Tools



Agent Services Options

Prompt Agents

- Author in portal or SDKs
- Foundry runs it for you
- No compute, containers, or packages

Hosted Agents

- Code your agent
- Package in container
- Run in Foundry
 - Managed endpoint
 - Scaling
 - Identity
 - Observability



Building Hosted Agents

- Bring your own code
- Protocol options
- Control your compute resources in Foundry
- Persist files and state
- Demo – Create a hosted agent



Extend agents with tools

Built-in tools

- Web Search
- Code Interpreter
- File Search
- Function Calling

Custom tools

- Model-Context-Protocol (MCP)
- Agent2Agent (A2A)
- OpenAPI



Toolboxes in Foundry

Build

- Select, configure and deploy tools in a toolbox
- Build once, reuse across agents

Consume

- Connect your agent to an MCP endpoint
- Access all tools in the toolbox

Demo

Create and deploy a toolbox in
Foundry





Manage agents at scale

- Microsoft Foundry Control Plane
 - Inventory
 - Observability
 - Tracing
 - Manage lifecycle events

Additional Resources



Foundry docs on
Microsoft Learn



Agent Services
docs



Azure SDKs



REST API
reference



Slides and demo
code



Thank you / Q&A

- Any questions?
- Contact me – alvin@alvinashcraft.com
- Bluesky –
<https://bsky.app/profile/alvinashcraft.com>

